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Spectral clustering

Jan Skalski

student record book number 320580

thesis supervisor
prof. PW, dr hab. Wojciech Matysiak

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Abstract

Spectral clustering

Spectral clustering is a relatively recent technique in the significant field of explanatory data analysis known as clustering. It has emerged as a powerful and versatile approach for identifying distinct clusters of similar observations. By leveraging only basic linear algebra concepts, spectral clustering algorithms typically avoid computational difficulties and yield satisfactory results even for very diverse data sets. This thesis provides a comprehensive explanation of the spectral clustering topic and the theoretical motivation behind it. The graph partitioning problem is introduced and elaborated upon as a natural and effective approach to clustering data represented in the form of a graph. Additionally, the structural properties of graphs are explored and emphasized as a foundational aspect of spectral clustering. Finally, the most common spectral clustering algorithms are presented, along with their functionality described in an intuitive manner using straightforward examples.

Keywords: clustering, spectral graph theory, graph partitioning

Streszczenie

Spektralna analiza skupień

Spektralna analiza skupień jest stosunkowo nową metodą w dziedzinie klasteryzacji danych. Wśród wielu innych algorytmów analizy skupień wyróżnia się jako bardzo efektywne i uniwersalne narzędzie, które zazwyczaj nie jest wymagające obliczeniowo oraz daje adekwatne wyniki nawet dla danych o bardzo nieregularnej strukturze. Niniejsza praca zawiera kompleksowe wyjaśnienie mechanizmów spektralnej analizy skupień, podparte wnikliwą teoretyczną motywacją. Wprowadzony i szczegółowo rozwinięty został temat reprezentacji danych w postaci grafowej, a także problem partycjonowania grafów w kontekście optymalnej klasteryzacji. Następnie przedstawione są konstrukcyjne własności grafów jako kluczowe aspekty odpowiadające za skuteczność klasteryzacji spektralnej. Na koniec zaprezentowane zostały podstawowe algorytmy spektralnej analizy skupień wraz z omówieniem ich działania z praktycznego punktu widzenia, wykorzystując w tym celu intuicyjne przykłady.

Słowa kluczowe: analiza skupień, spektralna teoria grafów, partycjonowanie grafu

Contents

1. Introduction

Data clustering is an essential practice for effective management of the vast amount of information available today. Its primary objective is to distinguish distinct subgroups within the data, commonly referred to as clusters, wherein observations share common characteristics or exhibit a certain degree of similarity to one another. This aim can be achieved through the implementation of appropriate algorithms with adjusted hyperparameters, frequently preceded by a certain level of data interpretation. As a form of unsupervised machine learning, clustering algorithms autonomously identify common patterns within the data without requiring any additional human contribution. They are particularly applied to determine regular and homogeneous regions within the data to recover its most fundamental properties, reduce excessive data distortion and retrieve the core information, or prepare it for further advanced data mining tasks that necessitate a more organized and streamlined data structure.

Not surprisingly, a variety of clustering algorithms exist, ranging from the widely-known k-means method and hierarchical clustering to the primary focus of this thesis, spectral clustering. While many other techniques typically pose significant limitations in handling diverse and demanding data, often failing to unveil its true nature, spectral clustering emerges as notably more methodical and versatile approach. Initially, it aims to represent the data structure and connections within it in the form of a graph, which is a natural and convenient choice. This way, regardless of their type or shape, observations can be seen as vertices, while information about the similarity between each pair is conveyed through the edges. However, this graph employment is not exceptional in the field of data clustering. What truly sets spectral clustering apart is its ability to leverage such representation and transform it effectively to capture the essence of the data's heterogeneity. Essentially, it achieves this goal by unraveling the constructional properties of graphs.

To more broadly illustrate its performance, the *spectral algorithm* only requires a certain *similarity measure* among observations as an input. It then follows a series of steps that ultimately embed the original arbitrary dataset into real points in a lower-dimensional Euclidean space, rendering the new embedded data structure more amenable for basic clustering methods, such as the k-means method. Eventually, in the final step, the clusters thus obtained are mapped back onto the initial data, completing the *spectral algorithm's* work. Given its comprehensive nature, the *spectral algorithm* is not as complex or computationally aggravating as it may seem. Indeed, it heavily relies solely on fundamental concepts from graph theory and linear algebra and its theoretical motivation is not challenging to acknowledge.

The primary objective of this thesis is to comprehensively demonstrate the entire process constituting the spectral clustering, elaborating upon the rationales established in von Luxburg (2007) and Bandeira et al. (2020) in an intelligible and more streamlined manner. To ensure clarity and consecutive order, thesis is divided into several parts.

The subsequent Chapter 2 introduces *similarity measures* and *similarity graphs* as indispensable elements of data clustering, enabling the clear representation of relationships within the data from both local and global perspectives.

The majority of the thesis is concentrated in Chapter 3, which delves into the mathematical underpinnings of spectral clustering. Its first section is dedicated to necessary preliminaries, providing the notational groundwork for further exploration of graph theory and linear algebra concepts. The second section examines special matrices known as *Graph Laplacians*, elucidating their key properties as meaningful characteristics of data structure, regarding the spectral clustering. The third section addresses fundamental tools for *graph partitioning* through various types of graph *cuts*, followed by framing the goal of effective clustering as a minimization task over specific graph *partitions*. The last yet most extensive section of Chapter 3 intricately explores the process of searching for a solution to the raised optimization problem, detailing the transitions and relaxations that ultimately lead to the analysis of the eigenvectors of certain *Graph Laplacians*.

Building upon the theoretical foundation established in the preceding chapters, the final Chapter 4 exhibits a step-by-step presentation of *spectral clustering algorithms*, complemented by practical explanations and visualizations derived from simple generated data examples. For this purpose, a straightforward implementation of the *spectral clustering algorithms* was created, equipped with certain additional attributes to facilitate the examination of the algorithms' successive steps and the structure of *Graph Laplacians*. It is available at: `https://github.com/klassik199/spectral-clustering-algorithm.git`

2. Similarity assessement

2.1. Similarity measures

To assign meaning to the clustering problem, it is crucial to initially determine the nature of similarity between observations in the dataset. By definition, the **similarity measure** should be a two-argument function defined over the data domain and return real non-negative values.

As similarity is inherently influenced by the dataset's type and applications, various forms of resemblance measures can be employed, all sharing the common goal of reflecting the genuine significance of the data. For instance, when data contains text objects that can be represented as real vectors, a convenient way to express similarity is through the *cosine similarity measure*, defined as the Euclidean dot product of two vectors. When the data has a grid structure, such as when it encodes an image, the preferred approach for a *similarity measure* is the *Manhattan distance*:

$$d_M(x, y) = \sum_i |x_i - y_i|.$$

In turn, when observations are strictly numerical or can be mapped as such, the most common approach to measuring resemblance is via the *Gaussian similarity function*:

$$s(x, y) = exp(\frac{\|x - y\|^2}{2\sigma^2}).$$

The parameter σ controls the sensitivity of the *similarity function* value to the distance between observations. For low values, points need to be very close to each other to be considered similar, and conversely, for higher values of σ, points can be farther apart and still receive high similarity.

2.2. Similarity graphs and adjacency matrices

Besides defining the *similarity measure*, another important aspect of preparing data for clustering is constructing the **similarity graph** and its corresponding square **adjacency matrix**, which binarily indicates whether the points on the graph are connected or not. For clarity purposes, it is assumed that the *adjacency matrix* has zeros along the main diagonal.

A *similarity graph* represents an anticipatory grouping of the data, based on a priori knowledge about the relationships within it. The appropriate choice of this graph is crucial for the success of the clustering algorithm, and it should be selected according to the local characteristics of the analyzed data. There are several approaches to determine the *similarity graph*:

- **k-nearest neighbors graph -** This graph connects each point only to the k points, for which it has the highest *similarity measure*. Since the nearest neighbors relation is not symmetric, it is possible that within two connected points, one may not be among the k-nearest neighbors of the other. This asymmetry does not occur in the mutual version of this *similarity graph*.
- **mutual k-nearest neighbors graph** -Within this *similarity graph*, an edge is formed between two points only if the first is among the k-nearest neighbors of the second and the second is among the k-nearest neighbors of the first, according to the *similarity measure*.
- **ϵ - neighborhood graph -** In this type of *similarity graph*, connected subgroups can be less balanced in terms of size, as each point is connected only to those points, for which the *similarity function* returns a value no greater than $\epsilon > 0$.
- **connected similarity graph -** When no additional details on the local relations within the data are provided, the most suitable yet more computationally demanding way of representing its structure is through a *connected similarity graph*, where each pair of points is connected.

3. Graph partitioning

After introducing *similarity measures* and *similarity graphs*, the concept of expressing data in the form of a graph can be outlined and expanded more precisely. In general, similarity is expressed as a continuous function. Therefore, to fully represent the data, it is reasonable to consider weighted graphs, where each edge carries a non-negative weight equal to the resemblance between the points it connects. This entails the need for a **weighted adjacency matrix**, wherein the entries corresponding to connected points on the *similarity graph* contain the value of their *similarity measure*.

With this data representation, the problem of detecting and grouping similar points can be expressed as dividing the set of vertices of the graph into disjoint, non-empty subsets, ensuring that the edges connecting vertices from different subsets have distinctly smaller weights than the edges between vertices within the same subset. Therefore, owing to this approach clustering arbitrary objects comes down to an optimization problem, which could be handled using standard linear algebra methods.

A detailed examination will be conducted on this task, exploring how spectral clustering advances towards the solution of the mentioned **graph partitioning** problem. However, for clarification purposes, some remarks on graph properties and notation that will be further used need to be introduced first.

3.1. Notation

Let $G = (V, E, W)$ be a weighted graph, where $V = \{1, 2, ..., n\} =: [n]$ is the set of vertices, $W = \{w_{ij}\}_{i,j \in V}$ is the *weighted adjacency matrix*, and $E = \{(i, j) : w_{ij} > 0,\ i, j \in V\}$ is the set of edges. Due to the fact that, in most cases similarity of the data points is regarded as a mutual and symmetric relationship, it is assumed that graph G is undirected, which means that each edge is bidirectional. Thus, it is clear to see that matrix W is symmetric, which is a crucial property to be emphasised later on.

Additionally, the non-empty subset of vertices $C \subseteq V$ is termed a **connected component** of G, if the subgraph induced by C is maximally connected, which can be denoted as:

$$\forall\, i, j \in C, \quad \forall\, l \in C^{\complement} := V \setminus C : \qquad w_{ij} > 0, \qquad \text{and} \qquad w_{il} = 0.$$

It is easy to observe that the union of all *connected components* within the graph G constitutes the entire set of vertices V. The number of all *connected components* of the graph is influenced by the type of *similarity graph* chosen. In the case of a *connected similarity graph*, there is only one *connected component* comprising the entire set of vertices V.

Now, to measure local significance, a **degree** of a vertex $i \in V$ is defined as:

$$deg(i) = \sum_{j \in V} w_{ij}.$$

Vector d stores the *degrees* of vertices, while the diagonal matrix D contains them along the main diagonal. Then for $i \in V$:

$$d_i = deg(i), \qquad D = diag(d).$$

The symbols $\sqrt{d}$ and $D^{-\frac{1}{2}}$ will denote a vector and a matrix satisfying:

$$\langle \sqrt{d}, \sqrt{d} \rangle = \langle d, \mathbb{1} \rangle, \qquad \text{and} \qquad D^{-\frac{1}{2}} D^{-\frac{1}{2}} = D^{-1}.$$

As weights of the edges are taken into consideration, two approaches for measuring the size of the subset $S \subseteq V$ can now be distinguished. First, the conventional one, that only counts the number of vertices within the subset, will be denoted as $|S|$. On the other hand, the second one, referred to as a **volume of the subset** provides a relevant knowledge on how dense (in terms of similarity) is the subset S in regard to the whole graph:

$$vol(S) = \sum_{i \in S} \sum_{j \in V} w_{ij} = \sum_{i \in S} deg(i).$$

The term **graph partition** relates to the division of the set of vertices V into distinct, non-empty subsets, whose union constitutes the entire set V. These subsets will be sometimes called the **components of a partition**. Explicitly, subsets $S_1, S_2, ..., S_k \subsetneq V$ $(2 \leqslant k \leqslant n)$ form a *partition* of G only if:

$$\forall i, j \in [k] : \quad S_i \cap S_j = \emptyset, \quad S_i \neq \emptyset, \quad \text{and} \quad \bigcup_{i \in [k]} S_i = V.$$

One can easily observe that *connected components* within the graph are an exceptional kind of *components* forming a *partition*, only if there exist at least two of them.

Regarding the vector notation, when multiple vectors appear and require additional indexing, their entries will be denoted with rounded brackets. For instance, $v_k(i)$ corresponds to the i-th entry of the k-th vector. The vector of zeros will be denoted as $\mathbb{0}$ and to associate a vector with a specific subset $S \subseteq V$, an indicator vector is introduced as follows:

$$\mathbb{1}_S \in \{0,1\}^n, \quad \mathbb{1}_S = \begin{cases} 1 & \text{if i} \in \text{S} \\ 0 & \text{otherwise} \end{cases}, \quad \text{and to simplify} \quad \mathbb{1} := \mathbb{1}_V.$$

The linearly dependent eigenvectors of the certain matrix will be referred to as the same vectors.

Now, concerning algebraic symbols, the dot product of two vectors $x, y \in \mathbb{R}^n$ and the Euclidean norm are designated respectively as:

$$\langle x, y \rangle = x^T y\ , \quad \|x\| = \langle x, x \rangle^{\frac{1}{2}}, \quad \text{and to shorten:} \quad x \perp y\ , \quad \text{if} \quad \langle x, y \rangle = 0.$$

Lastly, the identity matrix of size $n \times n$ is denoted as I, and the transpose of a matrix A as A^T.

3.2. Graph Laplacians

After covering the most important adopted notation, two useful matrices will now be introduced to enable further refinement of a *graph partitioning* problem. These matrices, known as **Graph Laplacians**, constitute the foundational element of study in the field of **spectral graph theory**, which delves into many structural properties of graphs and is comprehensively described in Chung (1997).

Definition 3.1 (Unnormalized Graph Laplacian). The term unnormalized *Graph Laplacian* associated with graph G describes a matrix:

$$L_G = D - W.$$

Definition 3.2 (Normalized Graph Laplacian). The normalized version of the *Graph Laplacian* is given by a matrix:

$$L_{NG} = D^{-\frac{1}{2}} L_G D^{-\frac{1}{2}} = I - D^{-\frac{1}{2}} W D^{-\frac{1}{2}}.$$

As will become evident later on, the eigenstructure of these matrices plays a crucial role in the *spectral clustering algorithms*. However, to justify this claim, their certain rather fundamental properties must first be distinguished. More elaborated insights into the *Graph Laplacians* and their other attributes can be found in Mohar (1991).

Lemma 3.3 (Properties of the Graph Laplacians).

1. L_G and L_{NG} are symmetric.
2. Let $1 \leqslant k \leqslant n$ be the number of *connected components* $C_1, ..., C_k$ within the graph G, then: $\mathbb{1}_{C_1}, ..., \mathbb{1}_{C_k}$ are the eigenvectors corresponding to the eigenvalue 0 of L_G, and $D^{\frac{1}{2}}\mathbb{1}_{C_1}, ..., D^{\frac{1}{2}}\mathbb{1}_{C_k}$ are the eigenvectors corresponding to the eigenvalue 0 of L_{NG}.
3. For both L_G and L_{NG}, the algebraic multiplicity of the eigenvalue 0 is equal to k.
4. $\forall x \in \mathbb{R}^n : \quad \langle x, L_G x\rangle = \sum_{i<j} w_{ij}(x_i - x_j)^2$.
5. Both L_G and L_{NG} are positive semi-definite and have real, non-negative eigenvalues.

Proof.
Point 1. is obvious from the symmetry of D and W. Then, after properly re-indexing the vertices of G, the *Graph Laplacians* L_G and L_{NG} can be expressed in the form of block diagonal matrices with each block corresponding to the distinct *connected component*. Clearly, by the construction each row in L_G sums up to 0, which, along with its block diagonal structure implies that the indicator vectors of the *connected components* are indeed the eigenvectors with the eigenvalue 0. Regarding the L_{NG}:

$$\forall\ i \in [k] : \quad L_{NG}(D^{\frac{1}{2}}\mathbb{1}_{C_i}) \ = \ D^{-\frac{1}{2}} L_G \mathbb{1}_{C_i} \ = \ \mathbb{0}.$$

This proves the second point, and consequently the third one, since in both scenarios mentioned indicator vectors are linearly independent of each other. As for the fourth statement, let $x \in \mathbb{R}^n$:

$$\begin{aligned}
\langle x, L_G x\rangle \ &= \ x^T(D-W)x \ = \ x^T D x - x^T W x \\
&= \sum_{i\in[n]} x_i^2 deg(i) - \sum_{i,j\in[n]} x_i x_j w_{ij} \ = \ \sum_{i,j\in[n]} x_i^2 w_{ij} - \sum_{i,j\in[n]} x_i x_j w_{ij} \\
&= \frac{1}{2}\Big[\sum_{i,j\in[n]} x_i^2 w_{ij} - \sum_{i,j\in[n]} 2x_i x_j w_{ij} + \sum_{i,j\in[n]} x_j^2 w_{ij}\Big] \\
&= \frac{1}{2}\sum_{i,j\in[n]} w_{ij}(x_i - x_j)^2 \ = \ \sum_{i<j} w_{ij}(x_i - x_j)^2.
\end{aligned}$$

Concerning the last property, the positive semi-definiteness of L_G is evident given point 4. and the non-negativity of the weights. Therefore, for the L_{NG}, for any $x \in \mathbb{R}^n$, it is sufficient to substitute $y = D^{-\frac{1}{2}}x$, then: $0 \leqslant \langle y, L_G y\rangle = \langle x, L_{NG} x\rangle$. The part, that eigenvalues of both L_G and L_{NG} are real follows from their symmetry and possession of real entries. Consequently, real eigenvectors exist for these matrices. Thus, considering this fact along with their positive semi-definiteness, it implies the non-negativity of eigenvalues. □

With all of that covered, most necessary *graph partitioning* tools can now be presented.

3.3. Graph cuts

Based on the introduced terminology, the clustering problem from the graph theory perspective could be redefined as the search for a *partition*, where edges connecting vertices from distinct *components* have possibly minimal weights compared to the edges within the same *component*. There naturally arises a need to specify a measure of the *partition's* quality. An intuitive way to achieve that is by introducing a **cut of a partition**.

Definition 3.4 (Cut of a partition). For a single, non-empty subset $S \subsetneq V$, *cut of a partition* $S, S^{\complement}$ (to simplify, sometimes it will be referred to as *cut* of S) is described as:

$$cut(S) = \sum_{i \in S} \sum_{j \in S^{\complement}} w_{ij},$$

and for multiple components $S_1, S_2, ..., S_k \subsetneq V$ $(k \geqslant 3)$ forming a *partition* of $G = (V, E, W)$:

$$cut(S_1, \dots, S_k) = \sum_{i \in [k]} cut(S_i).$$

From now on, proper cluster separation can be approached as a *partition*, that minimizes the *cut*. However, maintaining a certain balance between the *components* is a desirable goal, and simply optimizing the *cut* may not provide that. Indeed, this measure reaches its lowest value for *partitions* containing singleton with a vertex of the smallest *degree*. To address this issue, it is necessary to introduce more refined versions of the *cut*, that include the aforementioned objective.

3.3.1. Ratio cut

The initial idea is to somehow preserve balanced quantities of elements in *partition's components*. Generally, this can be accomplished using the **Ratio cut**, see Hagen and Kahng (1991).

Definition 3.5 (Ratio cut). For the subsets $S_1, S_2, \dots, S_k \subsetneq V$ $(k \geqslant 2)$ creating a *partition* of $G = (V, E, W)$, its *Ratio cut* is defined as follows:

$$Rcut(S_1, \dots, S_k) = \sum_{i \in [k]} \frac{cut(S_i)}{|S_i|}.$$

Assessing the *partition* using the *Ratio cut* measure, lower values will indicate, that there are no sparse *components*, unless their *cut* is really modest (which means that points inside them are highly dissimilar from the ones in different *components*). However, such an approach does not guarantee that the edges connecting vertices within a certain *component* in the *partition* have significantly larger weights compared to those extending outside of it.

3.3.2. Normalized cut

Mentioned disparity represents one of the principal objectives of *graph partitioning* and to large extent the **Normalized cut** (Shi and Malik, 2000) is feasible to overcome it.

Definition 3.6 (Normalized cut). For a partition $S_1, S_2, \dots, S_k \subsetneq V$ $(k \geqslant 2)$ of the graph $G = (V, E, W)$, its *Normalized cut* is given by:

$$Ncut(S_1, \dots, S_k) = \sum_{i \in [k]} \frac{cut(S_i)}{vol(S_i)}.$$

Normalizing *components cuts* with $vol(S_i)$ instead of $|S_i|$, leads to more balanced *partition* in terms of weight distribution. This allows them to be less numerous, but only if their weight density is really high. The *Normalized cut* effectively captures the flaws of most *graph partitions*, hence minimizing it with some constraints typically yields desirable and appropriate results.

3.3.3. Cheeger's cuts

A slightly different approach on the mentioned kinds of *cuts* is the **Cheeger's cut**. It involves considering the smaller of the respective sizes of the *component* and its complement as the normalizing factor. In the normalized version, this prevents from *partitions* with too large subsets combined of the portions with different weight densities. Its name originated from the relatively old study on the boundaries for smallest element in the spectrum of a *Laplacian* (Cheeger, 1971).

Definition 3.7 (Cheeger's cuts). For a two-*component partition* $S, S^{\complement} \subsetneq V$, its *Cheeger's cut* is defined as:

$$h(S) = \frac{cut(S)}{min\left\{|S|, |S^{\complement}|\right\}},$$

and the normalized version of the *Cheeger's cut* is given by:

$$h_N(S) = \frac{cut(S)}{min\left\{vol(S), vol(S^{\complement})\right\}}.$$

Not only is the *Cheeger's cut* approach more effective in measuring well-balanced *partitions*, but also its minimum can be determined quite accurately for a particular graph. It will be demonstrated and proved later on. A broader view on the *Cheeger's cut* and its properties can be found in Chang et al. (2015).

With the information on the most common *cuts of a partition* already provided, upon thorough examination, it will become apparent that minimizing them leads to the eigendecomposition of the respective *Graph Laplacians*.

3.4. Optimizing cuts using Graph Laplacians

To express the *cuts* through the *Graph Laplacians* terms effectively, leveraging the third point of the Lemma 3.3 will be crucial, as one might expect. To make subsequent steps more understandable, the *partitions* containing only two *components* will be considered primarily. However, to gain a broader concept on the *Graph Laplacians* properties, it is necessary to first provide a significant linear algebra theorem.

Theorem 3.8. Let L be real, symmetric matrix of size $n \times n$. Then eigenvectors of L form a orthonormal basis of $\mathbb{R}^n$.

The following proof would not be the most extensive one, but it will outline an intuitive yet satisfactory line of reasoning.

Proof. As was stated in the proof of the Lemma 3.3, eigenvalues of L are real due to its symmetry and possession of real entries. Therefore, for every eigenvalue there exist a real eigenvector. Let λ_1 denote an arbitrary eigenvalue of L and let v_1 be a real eigenvector corresponding to λ_1. Consider V_1 as the space spanned by vector v_1 and $V_1^{\perp}$ as its orthogonal complement:

$$V_1 = span(v_1) \quad \text{and} \quad V_1^{\perp} = \{u \in \mathbb{R}^n \ : \ \forall\, v \in V_1 \ \ u \perp v\}.$$

And obviously $dim(V_1^{\perp}) = dim(\mathbb{R}^n) - dim(V_1) = n - 1$ (this can be easily proved using Rank-nullity theorem). Now, observe that for any $u \in V_1^{\perp}$ and $v \in V_1$, the following holds:

$$\begin{aligned} &\langle Lu, v\rangle = \langle u, Lv\rangle = \lambda_1 \langle u, v\rangle = 0 && \text{by symmetry of } L, \qquad \Longrightarrow \\ &Lu \in V_1^{\perp} && \text{by arbitrariness of } v. \end{aligned}$$

Thus, arbitrariness of u implies, that $V_1^{\perp}$ is an invariant subspace of L, which means $LV_1^{\perp} \subseteq V_1^{\perp}$. Given that, every linear transformation has at least one eigenvalue over the complex numbers, and considering that L remains symmetric, while operating on $V_1^{\perp}$, there must exist a real eigenvalue λ_2 (not necessarily distinct from λ_1) and its associated real eigenvector $v_2 \in V_1^{\perp}$. Now, set $V_2 = span(v_1, v_2)$ and analogously one can show that $LV_2^{\perp} \subseteq V_2^{\perp}$, where $dim(V_2^{\perp}) = n - 2$. Following the same argumentation $n - 1$ times, successive eigenvalues and eigenvectors can be distinguished. And then, for $V_{n-1} = span(v_1, v_2, \ldots, v_{n-1})$, its orthogonal complement is one-dimensional, hence existence of the last eigenvector $v_n \in V_{n-1}^{\perp}$ is obvious. In this way, a set of n orthogonal (so also linear independent) eigenvectors of L was obtained, thus after normalizing them (by substituting $\frac{v_i}{\|v_i\|}$ for every $i \in [n]$), they create an orthonormal basis of $\mathbb{R}^n$. $\square$

The above theorem provides a meaningful knowledge, that the eigenvectors of the *Graph Laplacians* are perpendicular, which somehow depicts the direction of the following optimization process. But first, it is reasonable to represent the *cut* of a single *component* using the unnormalized *Graph Laplacian*, as it is quite straightforward. Indeed, for a non-empty subset $S \subsetneq V$:

$$cut(S) = \sum_{i \in S} \sum_{j \in S^{\complement}} w_{ij} = \sum_{i,j \in [n]} w_{ij}(\mathbb{1}_S(i) - \mathbb{1}_S(j))^2 = \langle \mathbb{1}_S, L_G \mathbb{1}_S \rangle.$$

However, such a representation does not facilitate the search for a minimum, as it still comes down to manually finding the appropriate subset, which is very challenging. An ultimate objective is to properly link a *component* S with a specific vector, for which the quadratic form on the right side can be minimized in some way.

Now regarding the refined versions of the *cut*, one can bring this goal closer by considering a discrete two-value vector $v \in \{a, b\}^n$, where $a, b \in \mathbb{R}$ are subject to logical constraints and v is associated with a subset S in a manner:

$$v_i = \begin{cases} a & \text{if } i \in S, \\ b & \text{otherwise.} \end{cases} \tag{3.1}$$

This along with the point 4. of Lemma 3.3 allows for a convenient representation:

$$\langle v, L_G v \rangle \;=\; \sum_{i \in S, j \in S^{\complement}} w_{ij}(a-b)^2 \;=\; cut(S)(a-b)^2. \tag{3.2}$$

The limitations on a feasible a and b values should incorporate the appropriate balance between linked component and its complement. Therefore, they will vary between the *Ratio cut* and the *Normalized cut* scenarios.

3.4.1. Minimizing Ratio cut of the two components

Since $cut(S) = cut(S^{\complement})$, the following holds:

$$Rcut(S, S^{\complement}) = \frac{cut(S)}{|S|} + \frac{cut(S^{\complement})}{|S^{\complement}|} = cut(S)\Big[\frac{1}{|S|} + \frac{1}{|S^{\complement}|}\Big] = cut(S)\frac{n}{|S||S^{\complement}|}. \tag{3.3}$$

So, (3.3) along with (3.2) give that a and b can be restricted as:

$$(a-b)^2 = \frac{n}{|S||S^{\complement}|}.$$

To establish the dependence of a and b, which involves the proportion between the sizes of S and its complement, an additional constraint is imposed:

$$a|S| + b|S^{\complement}| = 0.$$

Given those last two equalities, a and b can be determined:

$$b = -a\frac{|S|}{|S^{\complement}|} \qquad \Longrightarrow \qquad \left[a + a\frac{|S|}{|S^{\complement}|}\right]^2 = \frac{n}{|S||S^{\complement}|} \qquad \Longrightarrow \qquad a^2\frac{n^2}{|S^{\complement}|^2} = \frac{n}{|S||S^{\complement}|}.$$

Consequently:

$$a^2 = \frac{1}{n}\frac{|S^{\complement}|}{|S|} \qquad \Longrightarrow \qquad a = \sqrt{\frac{1}{n}\frac{|S^{\complement}|}{|S|}},$$
$$b^2 = \frac{1}{n}\frac{|S|}{|S^{\complement}|} \qquad \Longrightarrow \qquad b = -\sqrt{\frac{1}{n}\frac{|S|}{|S^{\complement}|}}.$$

Now, concerning the construction of v (3.1), it is easy to see that such requirements on a and b allow to restrict vector v in a more functional and general way:

$$\sum_{i\in[n]} v_i = |S|\sqrt{\frac{1}{n}\frac{|S^{\complement}|}{|S|}} - |S^{\complement}|\sqrt{\frac{1}{n}\frac{|S|}{|S^{\complement}|}} = 0 \qquad \Longrightarrow \qquad \langle v, \mathbb{1}\rangle = 0,$$
$$\sum_{i\in[n]} v_i^2 = \frac{|S^{\complement}|}{n} + \frac{|S|}{n} = 1 \qquad \Longrightarrow \qquad \|v\| = 1.$$

This finally contributes to the problem of minimizing the *Ratio cut* being reformulated in the terms of unnormalized *Graph Laplacian*:

$$\min_{\substack{S\subsetneq V\\ S\neq\emptyset}} Rcut(S, S^{\complement}) \quad = \min_{\substack{v\in\{a,b\}^n\\ v\perp\mathbb{1}\\ \|v\|=1}} \langle v, L_G v\rangle. \tag{3.4}$$

While such an equivalence represents a significant refinement of the problem, the discrete nature of the sought vectors makes finding the exact solution of (3.4) computationally very demanding. For an insightful discussion regarding this search, see Wagner and Wagner (1991). Fortunately, a method for its beneficial simplification exists and will be described shortly, following the minimization of the *Normalized cut*.

3.4.2. Minimizing Normalized cut of the two components

Since $vol(S) + vol(S^{\complement}) = vol(V)$, the *Normalized cut* can be expressed as:

$$Ncut(S, S^{\complement}) = \frac{cut(S)}{vol(S)} + \frac{cut(S^{\complement})}{vol(S^{\complement})} = cut(S)\frac{vol(V)}{vol(S)vol(S^{\complement})}. \tag{3.5}$$

Hence, by (3.2) combined with (3.5), a and b must now satisfy the following conditions:

$$(a-b)^2 = \frac{vol(V)}{vol(S)vol(S^{\complement})},$$
$$a\,vol(S) + b\,vol(S^{\complement}) = 0.$$

The second one is equivalent to:

$$\sum_{i\in S} v_i deg(i) + \sum_{i\in S^{\complement}} v_i deg(i) = \sum_{i\in[n]} v_i deg(i) = 0 \qquad \Longrightarrow \qquad \langle v, d\,\rangle = 0.$$

After similar transitions to previous ones, the a and b can be determined:

$$a^2 = \frac{1}{vol(V)}\frac{vol(S^{\complement})}{vol(S)} \qquad \Longrightarrow \qquad a = \sqrt{\frac{1}{vol(V)}\frac{vol(S^{\complement})}{vol(S)}},$$
$$b^2 = \frac{1}{vol(V)}\frac{vol(S)}{vol(S^{\complement})} \qquad \Longrightarrow \qquad b = -\sqrt{\frac{1}{vol(V)}\frac{vol(S)}{vol(S^{\complement})}}.$$

This implies that, the following must be true:

$$\begin{aligned}
&a^2 vol(S) + b^2 vol(S^{\complement}) = \frac{vol(S^{\complement})}{vol(V)} + \frac{vol(S)}{vol(V)} = 1 \qquad \Longrightarrow \\
&\sum_{i \in S} v_i^2 deg(i) + \sum_{i \in S^{\complement}} v_i^2 deg(i) = \sum_{i \in [n]} v_i^2 deg(i) = 1 \qquad \Longrightarrow \\
&\langle v, Dv \rangle = 1.
\end{aligned}$$

Therefore, the *Normalized cut* minimization can be expressed equivalently:

$$\min_{\substack{S \subsetneq V \\ S \neq \emptyset}} Ncut(S, S^{\complement}) \quad = \min_{\substack{v \in \{a,b\}^n \\ v \perp d \\ \langle v, Dv \rangle = 1}} \langle v, L_G v \rangle. \tag{3.6}$$

Substituting $z = D^{\frac{1}{2}} v$ in (3.6) eventually allows for the emergence of the quadratic form associated with the normalized *Graph Laplacian*:

$$\min_{\substack{S \subsetneq V \\ S \neq \emptyset}} Ncut(S, S^{\complement}) \quad = \min_{\substack{z \in \{a,b\}^n \\ z \perp \sqrt{d} \\ \|z\| = 1}} \langle z, L_{NG} z \rangle \,. \tag{3.7}$$

Likewise to the *Ratio cut*, in order to solve this optimization problem using standard methods, it is necessary to relax the discrete constraints on z in (3.7).

3.4.3. Spectral relaxation

A straightforward way of simplification involves considering real, continuous vectors rather than discrete ones. This results in the emergence of the lower bounds for the optimal *cuts*:

$$\min_{\substack{S \subsetneq V \\ S \neq \emptyset}} Rcut(S, S^{\complement}) \geqslant \min_{\substack{v \in \mathbb{R}^n \\ v \perp \mathbb{1} \\ \|v\| = 1}} \langle v, L_G v \rangle, \qquad \text{and} \qquad \min_{\substack{S \subsetneq V \\ S \neq \emptyset}} Ncut(S, S^{\complement}) \geqslant \min_{\substack{z \in \mathbb{R}^n \\ z \perp \sqrt{d} \\ \|z\| = 1}} \langle z, L_{NG} z \rangle. \tag{3.8}$$

One will eventually observe that this approach results in examining spectrum of the *Graph Laplacians*, and therefore, it will be referred to as a **spectral relaxation**. First, it is evident that such loosening forfeits the clear connection between vectors and *components* since now vector entries can assume arbitrary values. It will be demonstrated later, that such an issue is not as serious as it might initially seem. What is more significant though, is the fact that this *spectral relaxation* allows for explicit minimization solutions in both cases. Indeed, it will be precisely shown, that the relaxed expressions on the left-hand side and on the right-hand side in (3.8) are minimized for the eigenvectors corresponding to the second smallest eigenvalue of the *Graph Laplacian* L_G and the normalized *Graph Laplacian* L_{NG}, respectively.

Now, it is necessary to review some of the crucial *Graph Laplacians* properties, which will be further exploited. Primarily, by the point 5. in the Lemma 3.3 all eigenvalues of L_G and L_{NG} are non-negative. Additionally, the point 3. of Lemma 3.3 yields that $\mathbb{1}$ and $\sqrt{d}$ are the eigenvectors corresponding to the eigenvalue 0 of L_G and L_{NG} in a respective way, as:

$$\mathbb{1} = \sum_{i\in[k]} \mathbb{1}_{C_i} \quad \text{and} \quad \sqrt{d} = \sum_{i\in[k]} D^{\frac{1}{2}}\mathbb{1}_{C_i},$$

where $C_1, ..., C_k$ are the *connected components* in G. And as the *Graph Laplacians* are both symmetric, the Theorem 3.8 provides a crucial observation, that their eigenvectors form the orthonormal bases of $\mathbb{R}^n$.

From now on, the line of reasoning in this subsection remains the same, regardless of the type of the *cut*. Therefore it is sufficient to consider only the *Ratio cut* case.

Let $v^{(1)}, v^{(2)}, \dots, v^{(n)}$ represent the orthonormal eigenvectors of L_G corresponding to the eigenvalues $0 = \lambda^{(1)} \leqslant \lambda^{(2)} \leqslant ... \leqslant \lambda^{(n)}$ in a respective manner. Without loss of generality assume that $v^{(1)} = n^{-\frac{1}{2}}\mathbb{1}$ (eigenvector $\mathbb{1}$ corresponds to $\lambda^{(1)} = 0$). Also let v_2 denote a unit eigenvector associated with the second smallest eigenvalue $\lambda_2(L_G)$ ($\lambda_2(L_G) > 0$). Now, consider any vector $v \in \mathbb{R}^n$, that satisfies the constraints $v \perp \mathbb{1}$ and $\|v\| = 1$. As $v^{(1)}, \dots, v^{(n)}$ form a basis, then:

$$v = \sum_{i\in[n]} \alpha_i v^{(i)} \quad \text{for some} \quad \alpha_1, \dots, \alpha_n \in \mathbb{R}.$$

Since $v \perp v^{(1)}$ and $\|v\| = 1$:

$$\begin{aligned} 0 &= \langle v, v^{(1)} \rangle = \Big\langle \sum_{i\in[n]} \alpha_i v^{(i)}, v^{(1)} \Big\rangle = \sum_{i\in[n]} \alpha_i \langle v^{(i)}, v^{(1)} \rangle = \alpha_1, \\ 1 &= \langle v, v \rangle = \Big\langle \sum_{i\in[n]} \alpha_i v^{(i)}, \sum_{j\in[n]} \alpha_j v^{(j)} \Big\rangle = \sum_{i\in[n]} \sum_{j\in[n]} \alpha_i \alpha_j \langle v^{(i)}, v^{(j)} \rangle = \sum_{i\in[n]} \alpha_i^2. \end{aligned}$$

As a result:

$$\begin{aligned} \langle v, L_G v \rangle &= \Big\langle \sum_{i=2}^{n} \alpha_i v^{(i)}, L_G\Big(\sum_{j=2}^{n} \alpha_j v^{(j)}\Big) \Big\rangle = \sum_{i=2}^{n}\sum_{j=2}^{n} \alpha_i \alpha_j \langle v^{(i)}, L_G v^{(j)} \rangle \\ &= \sum_{i=2}^{n}\sum_{j=2}^{n} \alpha_i \alpha_j \lambda^{(j)} \langle v^{(i)}, v^{(j)} \rangle = \sum_{i=2}^{n} \alpha_i^2 \lambda^{(i)} \\ &\geqslant \sum_{i=2}^{n} \alpha_i^2 \lambda_2(L_G) = \lambda_2(L_G). \end{aligned}$$

And obviously $\langle v_2, L_G v_2 \rangle = \lambda_2(L_G)$. Due to arbitrariness of v, this implies that the searched minimum equals to the second smallest eigenvalue of L_G and it is reached for the unit eigenvector corresponding to it:

$$\min_{\substack{v\in\mathbb{R}^n \\ v\perp\mathbb{1} \\ \|v\|=1}} \langle v, L_G v \rangle \;=\; \langle v_2, L_G v_2 \rangle \;=\; \lambda_2(L_G). \tag{3.9}$$

By repeating analogous steps for L_{NG} with the respective notation, it is also acquired that:

$$\min_{\substack{z\in\mathbb{R}^n \\ z\perp\sqrt{d} \\ \|z\|=1}} \langle z, L_{NG} z \rangle \;=\; \langle z_2, L_{NG} z_2 \rangle \;=\; \lambda_2(L_{NG}). \tag{3.10}$$

Ultimately, by (3.8) along with (3.9) and (3.10), *spectral relaxation* provides the lower boundaries for optimal *cuts*:

$$\min_{\substack{S\subsetneq V \\ S\neq\emptyset}} Rcut(S, S^{\complement}) \;\geqslant\; \lambda_2(L_G), \qquad \text{and} \qquad \min_{\substack{S\subsetneq V \\ S\neq\emptyset}} Ncut(S, S^{\complement}) \;\geqslant\; \lambda_2(L_{NG}). \tag{3.11}$$

While this might not appear particularly helpful, vectors reaching such bounds play a crucial role in the spectral clustering algorithms. Being more specific, eigenvectors corresponding to the second smallest eigenvalues of the *Graph Laplacians* carry sufficient information about the connections between the vertices. To ensure that *spectral relaxation* provides desirable results, it is necessary to demonstrate that these eigenvectors can be effectively mapped onto a *partition* with a low *cut*.

3.4.4. Cheeger's inequality

The existence of such a mapping will be justified and elaborated upon, along with the upper bounds of the *cuts*. Since the main reasoning does not significantly differ between the *Ratio cut* and the *Normalized cut* cases, only the latter will be thoroughly described, as it is the more interesting one.

First, one can notice that such upper boundary for the *Normalized cut* can be obtained by bounding the normalized *Cheeger's cut*, as these two types of *cuts* are linked by pretty apparent inequalities. Indeed, for every non-empty subset $S \subsetneq V$:

$$\frac{1}{2} Ncut(S, S^{\complement}) \;\leqslant\; h_N(S) \;\leqslant\; Ncut(S, S^{\complement}). \tag{3.12}$$

As a reminder:

$$h_N(S) \;=\; \frac{cut(S)}{min\,\{vol(S), vol(S^{\complement})\}}.$$

Before introducing the significant theorem on the boundaries of the normalized *Cheeger's cut*, some remarks on the notation need to be outlined. An optimal *Cheeger's cut* value will be termed a **Cheeger's constant** (Mohar, 1989) and denoted as:

$$h_G := \min_{\substack{S \subsetneq V \\ S \neq \emptyset}} h_N(S). \tag{3.13}$$

Now, as previously let z_2 be the unit eigenvector corresponding to the second smallest eigenvalue $\lambda_2(L_{NG})$ of the normalized *Graph Laplacian*. As quadratic forms associated with the unnormalized *Graph Laplacian* have a convenient representation, a subtle substitution needs to be done to handily express $\lambda_2(L_{NG})$. Therefore, let $y = D^{-\frac{1}{2}} z_2$ and in this way:

$$\begin{aligned}
&\lambda_2(L_{NG}) \;=\; \langle\, z_2, L_{NG} z_2 \,\rangle \;=\; \langle\, y, L_G y \,\rangle, && \\
&\langle\, y, Dy \,\rangle \;=\; 1 && \text{by unit norm of } z_2, \\
&y \perp d && \text{because } z_2 \perp \sqrt{d}.
\end{aligned} \tag{3.14}$$

To bring some order to the entries of y, let $\forall\, i \in [n] :\ p_i$ denote the vertices, for which:

$$y\,(p_1) \;\leqslant\; y\,(p_2) \;\leqslant\; \ldots \;\leqslant\; y\,(p_n). \tag{3.15}$$

And for every $k \in [n]$, let $A_k \subseteq V$ represent the subset containing vertices related to the first k smallest values of y:

$$\forall\, k \in [n] \qquad A_k := \{p_1, \ldots, p_k\}.$$

Eventually, let a_G denote the optimal normalized *Cheeger's cut* restricted to the *partitions* based on these specific *components*:

$$a_G := \min_{k \in [n]} h_N(A_k). \tag{3.16}$$

Now, **Cheeger's inequality** can be presented as the fundamental theorem underpinning the effectiveness and accuracy of the *spectral clustering algorithms*. It was first established in the field of differential geometry (Cheeger, 1971), and later expanded to the following graph theory form in Alon and Milman (1985).

Theorem 3.9 (Cheeger's inequality).

$$\frac{1}{2}\lambda_2(L_{NG}) \;\leqslant\; h_G \;\leqslant\; a_G \;\leqslant\; \sqrt{2\lambda_2(L_{NG})}$$

Various approaches to prove this inequality are discussed in Chung (2007). The first of the four proofs in this article employs straightforward algebraic methods and will be presented here in a more streamlined and comprehensive form. To ensure the theorem's generality, a minor yet meaningful adjustment was made compared to the proof in Chung (2007). The primary mathematical focus of this thesis was dedicated to proving the *Cheeger's inequality*.

Proof. The left-hand side inequality is evident from (3.11) and (3.12) and the one in the middle is obvious, so the subsequent proof will solely focus on the right-hand side inequality. To remind: $y = D^{-\frac{1}{2}} z_2$, and it has properties described in (3.14) and (3.15). To gain broader knowledge on the signs of the y entries, it is necessary to somehow center this vector. Let m denote the first index, for which: $vol(A_m) \geqslant vol(A_m^{\complement})$. Explicitly:

$$\forall\, i < m : \quad vol(A_i) < vol(A_i^{\complement}), \qquad \text{and} \qquad \forall\, i \geqslant m : \quad vol(A_i) \geqslant vol(A_i^{\complement}). \tag{3.17}$$

Now, set centered vector x as:

$$x = y - y\,(p_m)\mathbb{1}.$$

It has certain useful properties following from (3.15):

$$\begin{aligned} &x\,(p_i) \leqslant 0 \qquad \forall\, i < m, \\ &x\,(p_m) = 0, \\ &x\,(p_i) \geqslant 0 \qquad \forall\, i > m, \\ &x\,(p_1) \leqslant x\,(p_2) \leqslant \ldots \leqslant x\,(p_n). \end{aligned} \tag{3.18}$$

This allows to easily distinguish positive and negative parts of the vector x. For every $i \in [n]$, their entries are introduced as follows:

$$x_-(p_i) = \begin{cases} -x\,(p_i) & \text{for } i < m, \\ 0 & \text{otherwise,} \end{cases} \qquad x_+(p_i) = \begin{cases} 0 & \text{for } i < m, \\ x\,(p_i) & \text{otherwise.} \end{cases}$$

Both vectors are obviously non-negative and their difference contributes to the entire vector: $x = x_+ - x_-$. Now, by point 4. of Lemma 3.3:

$$\langle\, x, L_G x \,\rangle = \sum_{i<j} w_{ij}(x_i - x_j)^2 = \sum_{i<j} w_{ij}(y_i - y_j)^2 = \langle\, y, L_G y \,\rangle.$$

In turn, $\langle\, x, Dx \,\rangle$ can be estimated from below by $\langle\, y, Dy \,\rangle$:

$$\begin{aligned} \langle\, x, Dx \,\rangle &= \langle\, y - y\,(p_m)\mathbb{1}, D(y - y\,(p_m)\mathbb{1}) \,\rangle \\ &= \langle\, y, Dy \,\rangle - 2y(p_m)\langle\, y, d \,\rangle + y\,(p_m)^2 \langle\, \mathbb{1}, d \,\rangle \\ &= \langle\, y, Dy \,\rangle + y\,(p_m)^2 \langle\, \mathbb{1}, d \,\rangle \qquad\qquad \text{as} \quad y \perp d, \\ &\geqslant \langle\, y, Dy \,\rangle. \end{aligned}$$

Then, by (3.14):

$$\lambda_2(L_{NG}) = \frac{\langle\, y, L_G y \,\rangle}{\langle\, y, Dy \,\rangle} \geqslant \frac{\langle\, x, L_G x \,\rangle}{\langle\, x, Dx \,\rangle}. \tag{3.19}$$

The right side can be further estimated using positive and negative parts of x. Indeed, since $x_-(i)x_+(i) \;=\; 0 \quad \forall\, i \in [n]$:

$$\langle\, x, Dx\,\rangle \;=\; \langle\, x_+, Dx_+\,\rangle \,+\, \langle\, x_-, Dx_-\,\rangle,$$

and, as x_+, x_- have non-negative entries:

$$\begin{aligned}
\langle\, x, L_G x\,\rangle \;&=\; \sum_{i<j} w_{ij}\Big[\big(x_+(i)-x_+(j)\big)-\big(x_-(i)-x_-(j)\big)\Big]^2\\
&=\; \sum_{i<j} w_{ij}\Big[\big(x_+(i)-x_+(j)\big)^2+2\big(x_+(i)x_-(j)+x_+(j)x_-(i)\big)+\big(x_-(i)-x_-(j)\big)^2\Big]\\
&\geqslant\; \sum_{i<j} w_{ij}\big(x_+(i)-x_+(j)\big)^2 \,+\, \sum_{i<j} w_{ij}\big(x_-(i)-x_-(j)\big)^2\\
&=\; \langle\, x_+, L_G x_+\,\rangle \,+\, \langle\, x_-, L_G x_-\,\rangle.
\end{aligned}$$

Therefore:

$$\frac{\langle\, x, L_G x\,\rangle}{\langle\, x, Dx\,\rangle} \;\geqslant\; \frac{\langle\, x_+, L_G x_+\,\rangle \,+\, \langle\, x_-, L_G x_-\,\rangle}{\langle\, x_+, Dx_+\,\rangle \,+\, \langle\, x_-, Dx_-\,\rangle} \;\geqslant\; min\left\{\frac{\langle\, x_+, L_G x_+\,\rangle}{\langle\, x_+, Dx_+\,\rangle}, \frac{\langle\, x_-, L_G x_-\,\rangle}{\langle\, x_-, Dx_-\,\rangle}\right\}.$$

The last inequality holds, as for every positive $a, b, c, d > 0$: $\quad \frac{a+b}{c+d} \;\geqslant\; min\left\{\frac{a}{c}, \frac{b}{d}\right\}$. Let firstly assume that:

$$\frac{\langle\, x_+, L_G x_+\,\rangle}{\langle\, x_+, Dx_+\,\rangle} \;\leqslant\; \frac{\langle\, x_-, L_G x_-\,\rangle}{\langle\, x_-, Dx_-\,\rangle}.$$

Hence:

$$\begin{aligned}
\frac{\langle\, x, L_G x\,\rangle}{\langle\, x, Dx\,\rangle} \;\geqslant\; \frac{\langle\, x_+, L_G x_+\,\rangle}{\langle\, x_+, Dx_+\,\rangle} \;&=\; \frac{\sum_{i<j} w_{ij}\big(x_+(i)-x_+(j)\big)^2}{\sum_{i\in[n]} x_+(i)^2 deg(i)}\\
&=\; \frac{\Big(\sum_{i<j} w_{ij}\big(x_+(i)-x_+(j)\big)^2\Big)^2}{\sum_{i\in[n]} x_+(i)^2 deg(i)\;\sum_{i<j} w_{ij}\big(x_+(i)+x_+(j)\big)^2} \;=:\; \frac{N}{D}.
\end{aligned}$$

Now, the numerator N can be estimated from below by Cauchy-Schwartz inequality:

$$N \geqslant \Big(\sum_{i<j} w_{ij}\big|x_+(i)^2 - x_+(j)^2\big|\Big)^2.$$

Since:

$$\forall\, i, j \in [n] : \qquad 2x_+(i)x_+(j) \leqslant x_+(i)^2 + x_+(j)^2, \qquad \text{and} \qquad \forall\, i \,\leqslant\, m : \quad x_+(p_i) \;=\; 0,$$

the term $\sum_{i<j} w_{ij}\big(x_+(i)+x_+(j)\big)^2$ can be bounded as:

$$\begin{aligned}
\sum_{i<j} w_{ij}\big(x_+(i)+x_+(j)\big)^2 \;&=\; \frac{1}{2}\sum_{i,j\in[n]} w_{ij}\big(x_+(i)+x_+(j)\big)^2 \;\leqslant\; 2\sum_{i,j\in[n]} w_{ij}x_+(i)^2\\
&=\; 2\sum_{i\in[n]} x_+(i)^2 deg(i) \;=\; 2\sum_{i>m} x_+(p_i)^2 deg(p_i).
\end{aligned}$$

So the denominator D is upper bounded as follows:

$$D \leqslant 2\Big(\sum_{i>m} x_+(p_i)^2 deg(p_i)\Big)^2.$$

This implies:

$$\frac{\langle x, L_G x\rangle}{\langle x, Dx\rangle} \geqslant \frac{\Big(\sum_{i<j} w_{ij}\big|x_+(i)^2 - x_+(j)^2\big|\Big)^2}{2\Big(\sum_{i>m} x_+(p_i)^2 deg(p_i)\Big)^2}. \tag{3.20}$$

Now, the crucial part involves cleverly estimating the term $\sum_{i<j} w_{ij}\big|x_+(i)^2 - x_+(j)^2\big|$ in the numerator on the right-hand side of (3.20). By (3.18), for every $i > j: \quad x_+(p_i) \geqslant x_+(p_j)$, therefore:

$$\begin{aligned}
\sum_{i<j} w_{ij}\big|x_+(i)^2 - x_+(j)^2\big| &= \frac{1}{2}\sum_{i,j\in[n]} w(p_i,p_j)\big|x_+(p_i)^2 - x_+(p_j)^2\big| \\
&= \sum_{i>j} w(p_i,p_j)\big|x_+(p_i)^2 - x_+(p_j)^2\big| \\
&= \sum_{i=1}^{n}\sum_{j=1}^{i-1} w(p_i,p_j)\Big(x_+(p_i)^2 - x_+(p_j)^2\Big) \\
&\Big[\text{since:} \quad x_+(p_i)^2 - x_+(p_j)^2 = \sum_{k=j}^{i-1}\Big(x_+(p_{k+1})^2 - x_+(p_k)^2\Big)\Big] \\
&= \sum_{i=1}^{n}\sum_{j=1}^{i-1}\sum_{k=j}^{i-1} w(p_i,p_j)\Big(x_+(p_{k+1})^2 - x_+(p_k)^2\Big) \\
&\left[\begin{cases} 1 \leqslant i \leqslant n \\ 1 \leqslant j \leqslant i-1 \\ j \leqslant k \leqslant i-1 \end{cases} \iff \begin{cases} 1 \leqslant k \leqslant n-1 \\ k+1 \leqslant i \leqslant n \\ 1 \leqslant j \leqslant k \end{cases}\right] \\
&= \sum_{k=1}^{n-1}\Big(x_+(p_{k+1})^2 - x_+(p_k)^2\Big)\sum_{i=k+1}^{n}\sum_{j=1}^{k} w(p_i,p_j) \\
&= \sum_{k=m}^{n-1}\Big(x_+(p_{k+1})^2 - x_+(p_k)^2\Big) cut(A_k).
\end{aligned}$$

Now, by (3.17) and the definition of a_G (3.16), for every $k \geqslant m$:

$$vol(A_k) \geqslant vol(A_k^{\complement}) \quad \Longrightarrow \quad cut(A_k) \geqslant a_G\, vol(A_k^{\complement}).$$

Thus:

$$\sum_{k=m}^{n-1}\Big(x_+(p_{k+1})^2 - x_+(p_k)^2\Big) cut(A_k) \geqslant a_G \sum_{k=m}^{n-1}\Big(x_+(p_{k+1})^2 - x_+(p_k)^2\Big) vol(A_k^{\complement}) =: (\bigstar).$$

By regrouping the sum:

$$(\bigstar) = a_G\Big[x_+(p_{m+1})^2\, vol(A_m^{\complement})$$

$$
\begin{aligned}
&\quad + \; x_+(p_{m+2})^2\, vol(A^{\subset}_{m+1}) \;-\; x_+(p_{m+1})^2\, vol(A^{\subset}_{m+1}) \\
&\quad + \; ... \\
&\quad + \; x_+(p_n)^2\, vol(A^{\subset}_{n-1}) \;-\; x_+(p_{n-1})^2\, vol(A^{\subset}_{n-1})\Big] \\
&= a_G\Big[x_+(p_{m+1})^2\Big(vol(A^{\subset}_{m}) \;-\; vol(A^{\subset}_{m+1})\Big) \\
&\quad + \; ... \\
&\quad + \; x_+(p_{n-1})^2\Big(vol(A^{\subset}_{n-2}) \;-\; vol(A^{\subset}_{n-1})\Big) \\
&\quad + \; x_+(p_n)^2\, vol(A^{\subset}_{n-1})\Big] \\
&= a_G \sum_{k>m} x_+(p_k)^2\, deg(p_k).
\end{aligned}
$$

Therefore, the term $\sum_{i<j} w_{ij}\big|x_+(i)^2 - x_+(j)^2\big|$ can be bounded from below as:

$$
\begin{aligned}
\sum_{i<j} w_{ij}\big|x_+(i)^2 - x_+(j)^2\big| \;&=\; \sum_{k=m}^{n-1}\Big(x_+(p_{k+1})^2 - x_+(p_k)^2\Big) cut(A_k) \\
&\geqslant\; a_G \sum_{k>m} x_+(p_k)^2\, deg(p_k).
\end{aligned}
$$

And eventually, by combining (3.19) with (3.20):

$$
\lambda_2(L_{NG}) \;\geqslant\; \frac{\langle\, x, L_G x\,\rangle}{\langle\, x, Dx\,\rangle} \;\geqslant\; \frac{a_G^2}{2}.
$$

Now, to sustain generality it is necessary to consider the case when:

$$
\frac{\langle\, x_-, L_G x_-\,\rangle}{\langle\, x_-, Dx_-\,\rangle} \;\leqslant\; \frac{\langle\, x_+, L_G x_+\,\rangle}{\langle\, x_+, Dx_+\,\rangle}.
$$

Through the same steps the analogous inequality to (3.20) is attained:

$$
\lambda_2(L_{NG}) \;\geqslant\; \frac{\langle\, x, L_G x\,\rangle}{\langle\, x, Dx\,\rangle} \;\geqslant\; \frac{\Big(\sum_{i<j} w_{ij}\big|x_-(i)^2 - x_-(j)^2\big|\Big)^2}{2\Big(\sum_{i<m} x_-(p_i)^2 deg(p_i)\Big)^2}.
$$

This time, for every $i \,<\, j:\quad x_-(p_i) \,\geqslant\, x_-(p_j),\;$ and for every $i \,\geqslant\, m:\quad x_-(p_i) = 0$, hence:

$$
\begin{aligned}
\sum_{i<j} w_{ij}\big|x_-(i)^2 - x_-(j)^2\big| \;&=\; \sum_{i<j} w(p_i,p_j)\big|x_-(p_i)^2 - x_-(p_j)^2\big| \\
&= \sum_{i=1}^{n}\sum_{j=i+1}^{n} w(p_i,p_j)\Big(x_-(p_i)^2 - x_-(p_j)^2\Big) \\
&\Big[\text{since:}\quad x_-(p_i)^2 - x_-(p_j)^2 \;=\; \sum_{k=i}^{j-1}\Big(x_-(p_k)^2 - x_-(p_{k+1})^2\Big)\Big] \\
&= \sum_{i=1}^{n}\sum_{j=i+1}^{n}\sum_{k=i}^{j-1} w(p_i,p_j)\Big(x_-(p_k)^2 - x_-(p_{k+1})^2\Big)
\end{aligned}
$$

$$
\begin{aligned}
&= \sum_{k=1}^{m-1}\Big(x_-(p_k)^2 - x_-(p_{k+1})^2\Big)\,cut(A_k) \\
&\geqslant a_G \sum_{k=1}^{m-1}\Big(x_-(p_k)^2 - x_-(p_{k+1})^2\Big)\,vol(A_k) \qquad\qquad \text{by (3.17)} \\
&= a_G\Big[x_-(p_1)^2\,vol(A_1) \;+\; \sum_{k=2}^{m-1} x_-(p_k)^2\Big(vol(A_k) \;-\; vol(A_{k-1})\Big)\Big] \\
&= a_G \sum_{k<m} x_-(p_k)^2\,deg(p_k).
\end{aligned}
$$

Therefore, the same boundary for $\lambda_2(L_{NG})$ is established which leads to the theorem's last inequality:

$$a_G \;\leqslant\; \sqrt{2\lambda_2(L_{NG})}.$$

Finally, this concludes the proof:

$$\frac{1}{2}\lambda_2(L_{NG}) \;\leqslant\; h_G \;\leqslant\; a_G \;\leqslant\; \sqrt{2\lambda_2(L_{NG})}.$$

□

There are some evident yet meaningful conclusions derived from the *Cheeger's inequality* along with the inequalities (3.11) and (3.12).

Corollary 3.10 (Outcomes of the Cheeger's inequality). Let $A_{opt} \in \{A_1, \dots, A_n\}$ denote the subset, for which a_G is achieved and let $S_{opt} \subsetneq V, S \neq \emptyset$ represent the subset that minimizes the *Normalized cut*. Then:

$$Ncut(S_{opt}, S_{opt}^{\complement}) \;\leqslant\; Ncut(A_{opt}, A_{opt}^{\complement}) \;\leqslant\; \sqrt{8\lambda_2(L_{NG})} \;\leqslant\; \sqrt{8Ncut(S_{opt}, S_{opt}^{\complement})},$$

and there exist a threshold $\tau \in \mathbb{R}$, such that:

$$A_{opt} = \{i \in [n] :\;\; y_i \;\leqslant\; \tau\}, \qquad \text{where} \quad y = D^{-\frac{1}{2}} z_2.$$

These conclusions yield a mapping of the eigenvector z_2 corresponding to $\lambda_2(L_{NG})$ into the *component* A_{opt}, whose *Normalized cut* is suboptimal by a factor of the square root of the optimum. In fact, to find the subset A_{opt}, it is sufficient to try different tresholds or to simply execute the 2-means clustering method to the entries of y. As all of this also applies to the *Ratio cut* case in a respective way, such knowledge constitutes the motivation for the *spectral clustering algorithms*, which will be further discussed in the next chapter. Before covering that, however, it is necessary to eventually consider the general case of optimizing the *Ratio cut* and the *Normalized cut* of the *partitions* with an arbitrary number of *components*.

3.4.5. Minimizing cuts of the multiple components

Assume that $S_1, \dots, S_k \subsetneq V$ $(3 \leqslant k \leqslant n)$ form a *partition* of G. Each of these *components* can be represented by a specific vector in a similar manner as before. Therefore, for every $i \in [k]$, $j \in [n]$, let $x_i \in \{a_i, 0\}^n$ $(a_i \neq 0)$ be the vector associated with a subset S_i as follows:

$$x_i(j) = \begin{cases} a_i & \text{if } j \in S_i, \\ 0 & \text{otherwise.} \end{cases} \tag{3.21}$$

Then, by point 4. of Lemma 3.3, for every $i \in [k]$:

$$\langle\, x_i, L_G x_i \,\rangle \;=\; cut(S_i) a_i^2. \tag{3.22}$$

Firstly, to represent the *Ratio cut* of $S_1, .., S_k$ in the terms of the unnormalized *Graph Laplacian*, a_i must satisfy:

$$\forall i \in [k] \quad a_i \;=\; \frac{1}{\sqrt{|S_i|}}.$$

In this way:

$$Rcut(S_1, \dots, S_k) \;=\; \sum_{i\in[k]} \frac{cut(S_k)}{|S_k|} \;=\; \sum_{i\in[k]} \langle\, x_i, L_G x_i \,\rangle \;=\; tr(X^T L_G X),$$

where X is the matrix containing vectors $x_1, \dots, x_k$ as columns. Now, by construction in (3.21) it is easy to acknowledge that $x_1, \dots, x_k$ are orthonormal, which can be equivalently expressed as: $X^T X = I$. Hence, the task of minimizing the *Ratio cut* of a *partition* is parallel to minimizing the trace of the matrix $X^T L_G X$ under certain constraints:

$$\min_{S_1,\dots,S_k \subsetneq V} Rcut(S_1, ..., S_k) \;=\; \min_{\substack{x_1,\dots,x_k \in \{a,0\}^n \\ X^T X = I}} tr(X^T L_G X). \tag{3.23}$$

Unsurprisingly, such discrete restrictions on the vectors $x_1, \dots, x_k$ do not sufficiently alleviate the difficulty in solving this problem. Therefore, as in the two *component* case, it is necessary to relax the constraints in (3.23) by allowing continuous entries. This leads to a lower bound for optimal *Ratio cut*:

$$\min_{S_1,\dots,S_k \subsetneq V} Rcut(S_1, ..., S_k) \;\geqslant\; \min_{\substack{x_1,\dots,x_k \in \mathbb{R}^n \\ X^T X = I}} tr(X^T L_G X).$$

As will be now presented, the right side of the above inequality has the explicit solution.

Let $v_1, ...v_n$ be the orthonormal eigenvectors of L_G corresponding to the eigenvalues: $\lambda_1(L_G) \leqslant ... \leqslant \lambda_n(L_G)$, respectively. In turn, let $\hat{v_1}, ...\hat{v_k}$ denote the orthonormal eigenvectors of the symmetric matrix $\widehat{L_G} = X^T L_G X \in \mathbb{R}^{k\times k}$ corresponding to the eigenvalues $\hat{\lambda_1} \leqslant ... \leqslant \hat{\lambda_k} \in \mathbb{R}$. For every $m \in [k]$ consider $\widehat{V_m} = span(\hat{v_1}, ..., \hat{v_m}) \subseteq \mathbb{R}^k$, and $X\widehat{V_m} = span(X\hat{v_1}, ..., X\hat{v_m}) \subseteq \mathbb{R}^n$.

Since $X^T X = I$, $X\widehat{V_m}$ is m-dimensional subspace of $\mathbb{R}^n$. Then, for fixed $m \in [k]$ and any $\hat{v} \in \widehat{V_m}$, $\|\hat{v}\| = 1$ there exist $\alpha_1, ..., \alpha_m \in \mathbb{R}$, such that $\hat{v} = \sum_{i \in [m]} \alpha_i \hat{v_i}$, and $\sum_{i \in [m]} \alpha_i^2 = 1$. This, along with the orthonormality of $\hat{v_1}, ..., \hat{v_k}$, implies that:

$$\langle \hat{v}, \widehat{L_G} \hat{v} \rangle \ = \ \sum_{i \in [m]} \alpha_i^2 \hat{\lambda_i} \ \leqslant \ \hat{\lambda_m}.$$

And obviously $\langle \hat{v_m}, \widehat{L_G} \hat{v_m} \rangle \ = \ \hat{\lambda_m}$, then:

$$\hat{\lambda_m} \ = \max_{\substack{\hat{v} \in \widehat{V_m} \\ \|\hat{v}\| = 1}} \ \langle \hat{v}, \widehat{L_G} \hat{v} \rangle \ = \max_{\substack{\hat{v} \in \widehat{V_m} \\ \|\hat{v}\| = 1}} \ \langle X\hat{v}, L_G X \hat{v} \rangle.$$

Substituting $v = X\hat{v}$ retains the norm of the vector, as $X^T X = I$. This allows for the equivalent representation of the above maximum:

$$\hat{\lambda_m} \ = \max_{\substack{\hat{v} \in \widehat{V_m} \\ \|\hat{v}\| = 1}} \ \langle X\hat{v}, L_G X \hat{v} \rangle \ = \max_{\substack{v \in X\widehat{V_m} \\ \|v\| = 1}} \langle v, L_G v \rangle. \tag{3.24}$$

Since $X\widehat{V_m} \cap span(v_m, v_{m+1}, ..., v_n) \neq \emptyset$, there exist $v^* \in X\widehat{V_m}$, $\|v^*\| = 1$, such that $v^* = \sum_{i=m}^n \beta_i v_i$, and $\sum_{i=m}^n \beta_i^2 = 1$ for some $\beta_m, ..., \beta_n \in \mathbb{R}$. Then:

$$\langle v^*, L_G v^* \rangle \ = \ \sum_{i=m}^{n} \beta_i^2 \lambda_i(L_G) \ \geqslant \ \lambda_m(L_G).$$

Therefore, combining this with (3.24) gives the following:

$$\hat{\lambda_m} \ = \max_{\substack{v \in X\widehat{V_m} \\ \|v\| = 1}} \langle v, L_G v \rangle \ \geqslant \ \langle v^*, L_G v^* \rangle \ \geqslant \ \lambda_m(L_G).$$

Eventually, arbitrariness of $m \in [k]$ yields that:

$$tr(X^T L_G X) \ = \ \sum_{i \in [k]} \hat{\lambda_i} \ \geqslant \ \sum_{i \in [k]} \lambda_i(L_G).$$

Such lower bound for the trace of $X^T L_G X$ is clearly achieved by the $v_1, ..., v_k$ as the columns of X, which concludes:

$$\min_{\substack{x_1, ..., x_k \in \mathbb{R}^n \\ X^T X = I}} tr(X^T L_G X) \ = \ \sum_{i \in [k]} \langle v_i, L_G v_i \rangle \ = \ \sum_{i \in [k]} \lambda_i(L_G). \tag{3.25}$$

Thus, the orthonormal eigenvectors corresponding to the first k smallest eigenvalues of L_G (counted with the algebraic multiplicities) minimize the relaxed problem, which also means that they carry enough information about the relationships within vertices.

Now, considering the *Normalized cut* of a *partition* $S_1, ..., S_k$, it is easy to see that this time a_i must be equal to:

$$\forall i \in [k] \quad a_i \ = \ \frac{1}{\sqrt{vol(S_i)}}.$$

Leveraging the (3.22), the following representation is achieved for the vectors x_i defined in the same way as in (3.21):

$$Ncut(S_1, \dots, S_k) \;=\; \sum_{i \in [k]} \frac{cut(S_k)}{vol(S_i)} \;=\; \sum_{i \in [k]} \langle\, x_i, L_G x_i \,\rangle.$$

Substituting $y_i = D^{\frac{1}{2}} x_i \;\; \forall\, i \in [k]$, and $Y = D^{\frac{1}{2}} X$, the normalized *Graph Laplacian* emerges:

$$\sum_{i \in [k]} \langle\, x_i, L_G x_i \,\rangle \;=\; \sum_{i \in [k]} \langle y_i, L_{NG} y_i \rangle \;=\; tr(Y^T L_{NG} Y).$$

By construction of $x_1, ..., x_k$ in (3.21), it is easy to see that $y_1, ..., y_k$ are orthonormal as well. Hence, $Y^T Y = I$ and the *Normalized cut* minimization can be expressed as follows:

$$\min_{S_1, \dots, S_k \subsetneq V} Ncut(S_1, ..., S_k) \;=\; \min_{\substack{y_1, ..., y_k \in \{a, 0\}^n \\ Y^T Y = I}} tr(Y^T L_{NG} Y). \tag{3.26}$$

After *spectral relaxation*, the right-hand side of (3.26) is solved by the eigenvectors $z_1, ..., z_k$ corresponding to the first k smallest eigenvalues $\lambda_1(L_{NG}) \leqslant ... \leqslant \lambda_k(L_{NG})$ of the normalized *Graph Laplacian*:

$$\min_{\substack{y_1, ..., y_k \in \mathbb{R}^n \\ Y^T Y = I}} tr(Y^T L_{NG} Y) \;=\; \sum_{i \in [k]} \langle z_i, L_{NG} z_i \rangle \;=\; \sum_{i \in [k]} \lambda_i(L_{NG}). \tag{3.27}$$

Therefore, (3.25) and (3.27) yield the final lower bounds for the optimal *cuts*:

$$\min_{S_1, \dots, S_k \subsetneq V} Rcut(S_1, ..., S_k) \;\geqslant\; \sum_{i \in [k]} \lambda_i(L_G), \qquad \text{and} \qquad \min_{S_1, \dots, S_k \subsetneq V} Ncut(S_1, ..., S_k) \;\geqslant\; \sum_{i \in [k]} \lambda_i(L_{NG}). \tag{3.28}$$

Eventually, equivalents of *Cheeger's constant* and *Cheeger's inequality* exist in the case of $k \geqslant 3$ *components*. They are analogous between the two *cuts*, therefore they will only be presented for the *Normalized cut* scenario. The ***k*-way expansion constant** (Lee et al., 2011) is defined as:

$$\rho_G(k) \;=\; \min_{\substack{S_1, \dots, S_k \subsetneq V \\ S_i \neq \emptyset,\;\; S_i \cap S_j = \emptyset \\ \forall i, j \in [k]}} \; \max_{l \in [k]} \; Ncut(S_l, S_l^{\complement}), \tag{3.29}$$

and the **generalised Cheeger's inequality** has the following form:

$$\frac{\lambda_k(L_{NG})}{2} \;\leqslant\; \rho_G(k) \;\leqslant\; O(k^2) \sqrt{\lambda_k(L_{NG})}. \tag{3.30}$$

Additionally, an exhaustive analysis on how thresholding the entries of the mentioned eigenvectors leads to nearly optimal *partitions*, can be found in Lee et al. (2011), along with the proof of inequality (3.29). In fact, k-means clustering the k-dimensional points indicated by the distinct coordinates of the eigenvectors $v_1, ..., v_k$ is enough to obtain the *partition* with a low *Ratio cut*. Similarly, to achieve a *partition* with a low *Normalized cut*, it is sufficient to cluster the k-dimensional points given by the distinct entries of the vectors $D^{-\frac{1}{2}} z_1, ..., D^{-\frac{1}{2}} z_k$ using the k-means method. The rationale behind why the k-means algorithm yields good results in this case will be presented in the last section of the following chapter.

4. Spectral clustering algorithms

After comprehensive mathematical motivation, two most important *spectral clustering algorithms* can finally be demonstrated. Assume that $x_1, ..., x_n$ denote the data points of an arbitrary type and dimension. For every $i, j \in [n], i \neq j$ let s_{ij} be the similarity between points x_i, x_j measured by some non-negative and symmetric function. The objective is to properly cluster these points into $k \geqslant 2$ distinct groups with similar points within them. The *spectral algorithms* are outlined as follows:

4.1. Unnormalized spectral algorithm

Unnormalized spectral clustering algorithm

1. Given the similarities s_{ij}, determine the similarity graph and its weighted adjacency matrix $W \in \mathbb{R}^{n \times n}$.
2. Compute the degree of each point and the diagonal matrix $D \in \mathbb{R}^{n \times n}$.
3. Construct the unnormalized Graph Laplacian L_G based on W and D as was stated in the section 3.2.
4. Compute the k-1 eigenvalues of L_G from the second smallest one up to the k-th smallest one, along with their corresponding eigenvectors $v_2, ..., v_k$, respectively.
5. Based on the distinct coordinates of the eigenvectors $v_2, ..., v_k$ construct the k-1-dimensional points $y_1, ..., y_n$, where $y_i(j-1) = v_j(i)$ for every $i = 1, ..., n$ and $j = 2, ..., k$.
6. Cluster the points $y_1, ..., y_n$ into k groups $A_1, ..., A_k$ using the k-means clustering method.
7. Return the clusters $C_1, ..., C_k$, where $C_j = \{x_i : y_i \in A_j\}$ for every $j = 1, ..., k$.

Considering the case of the multiple clusters ($k > 2$), one can notice a difference in the number of the eigenvectors, that need to be calculated, between the previous chapter reasoning and the above algorithm. This results from the fact, that the eigenvector $v_1 = \mathbb{1}$ corresponding to the eigenvalue 0 of L_G is constant, which implies it has no effect on the distances within the points $y_1, ..., y_n$. Therefore, to simplify the computation and unify the algorithm it can be omitted.

4.2. Normalized spectral algorithm

Normalized spectral clustering algorithm

1. Given the similarities s_{ij}, determine the similarity graph and its weighted adjacency matrix $W \in \mathbb{R}^{n \times n}$.
2. Compute the degree of each point and the diagonal matrix $D \in \mathbb{R}^{n \times n}$.
3. Construct the normalized Graph Laplacian L_{NG} based on W and D as was stated in the section 3.2.
4. Compute the k-1 eigenvalues of L_{NG} from the second smallest one up to the k-th smallest one, along with their corresponding eigenvectors $z_2, ..., z_k$, respectively.
5. Transform the eigenvectors $z_2, ..., z_k$ into $D^{-\frac{1}{2}} z_2, ..., D^{-\frac{1}{2}} z_k$.
6. Based on the distinct coordinates of the vectors $D^{-\frac{1}{2}} z_2, ..., D^{-\frac{1}{2}} z_k$ construct the k-1-dimensional points $y_1, ..., y_n$, where $y_i(j-1) = D^{-\frac{1}{2}} z_j(i)$ for every $i = 1, ..., n$ and $j = 2, ..., k$.
7. Cluster the points $y_1, ..., y_n$ into k groups $A_1, ..., A_k$ using the k-means clustering method.
8. Return the clusters $C_1, ..., C_k$, where $C_j = \{x_i : y_i \in A_j\}$ for every $j = 1, ..., k$.

To cover the lack of the eigenvector z_1 in the point 4. for the case of $k > 2$, same explanation as previously can be applied, as $z_1 = \sqrt{d}$, then $D^{-\frac{1}{2}} z_1 = \mathbb{1}$ is constant and does not carry any additional information on the relationships within the points $y_1, ..., y_n$.

Subsequent section will aim to elucidate the concept of the *spectral clustering algorithms* in a more intuitive and clarified manner. Through practical examples, its effect and advantages will be outlined and compared to the other clustering methods.

4.3. Functioning of the algorithms illustrated with examples

Using intentionally generated data, two straightforward examples will be presented in order to demonstrate the functionality and effectiveness of the *spectral algorithms*. To measure the resemblance within the data points, they will both employ the *Gaussian similarity function* with different hyperparameter σ, as it is the most common and versatile choice for the strictly numerical data. The *weighted adjacency matrix* used in the examples will be based on a *connected similarity graph*. Consequently, the graphs representing the analyzed data will consist of a single *connected component*.

The first example is derived from a dataset with 1000 one-dimensional points created from three normal samples X_1, X_2, X_3, each with distinct mean: $\mu_{X_1} = 25$, $\mu_{X_2} = 45$, $\mu_{X_3} = 55$, standard deviation: $sd_{X_1} = 8$, $sd_{X_2} = 3$, $sd_{X_3} = 1$ and size: $|X_1| = 600$, $|X_2| = 300$, $X_3 = 100$. Employed *Gaussian similarity function* has a standard specification with $\sigma = 1$.

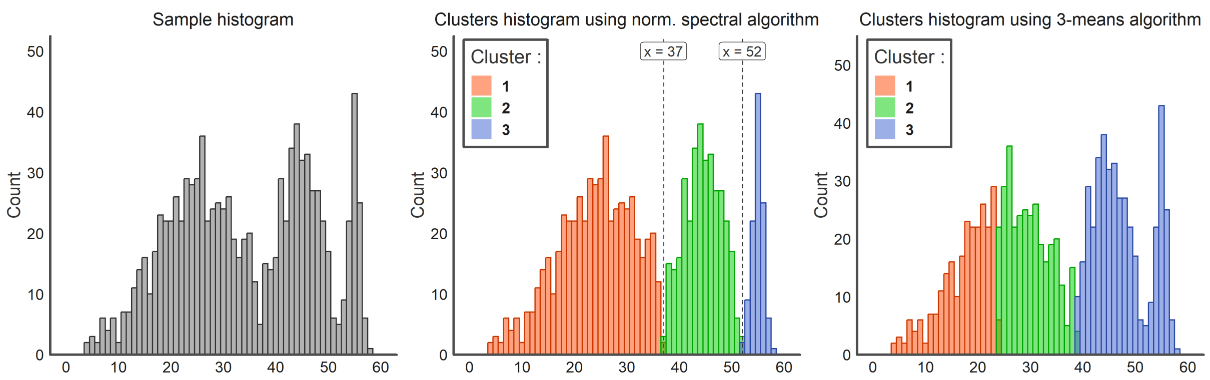

Figure 4.1: Sample histogram on the left, partitioned histogram via *normalized spectral algorithm* for 3 clusters in the middle (vertical, dashed lines approximately indicate the boundaries between neighboring clusters), partitioned histogram via 3-means clustering on the right.

As depicted in the histograms above, the *normalized spectral algorithm* identifies the observations from distinct distributions almost perfectly, whereas employing only the k-means formula yields results far from satisfactory. This is because, the latter method tends to capture different clusters properly solely when they are spherical and possess comparable densities and sizes. In this instance, such a condition is not fulfilled, making this basic method alone insufficient. The necessity for a spectral algorithm emerges here, as it effectively transforms arbitrarily diverse data into more streamlined and uniformly distributed, thereby facilitating the accurate detection of clusters by the k-means method in its final stage.

The next figure demonstrates, that the mentioned transformation originates from the form of certain vectors associated with the *Graph Laplacians*.

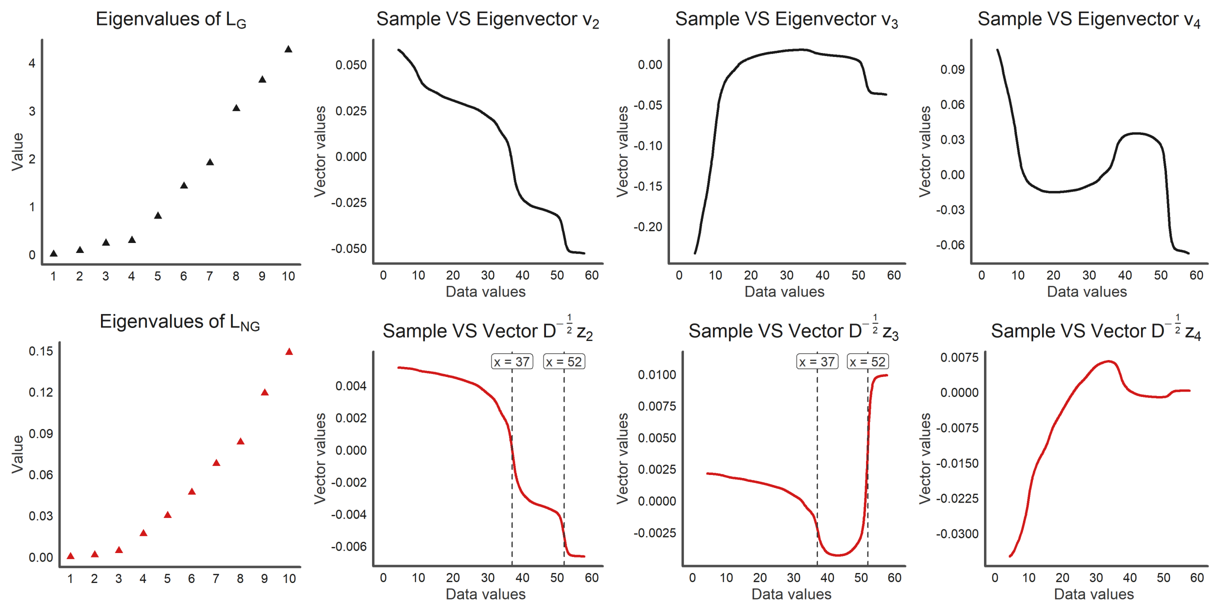

Figure 4.2: First column contains the plots of the 10 smallest eigenvalues of the *Graph Laplacian* L_G and its normalized version L_{NG}, respectively. Other plots represent the entries of the eigenvectors v_2, v_3, v_4 of the L_G in the first row and the vectors $D^{-\frac{1}{2}}z_2, D^{-\frac{1}{2}}z_3, D^{-\frac{1}{2}}z_4$ in the second one, plotted against the sample values (z_2, z_3, z_4 are the eigenvectors of the L_{NG}) .

Now, one can easily observe that the coordinates of the vectors $D^{-\frac{1}{2}}z_2$ and $D^{-\frac{1}{2}}z_3$ reveal the same three-part pattern, which is indicated by the vertical, dashed lines. These two vectors provide the effective partitioning of the data into three clusters. A threshold of approximately 0 distinguish the first cluster from the remaining two in both vectors. And to separate the third cluster from the second one, it is necessary to establish a threshold of around -0.005 in the vector $D^{-\frac{1}{2}}z_2$ and around 0.005 in the $D^{-\frac{1}{2}}z_3$.

Additionally, there is a noticeable difference in the curve's slope dividing the first and second clusters compared to the one between the second and third clusters, which resembles the sample histogram, as the latter pair of the clusters is more detached from each other in comparison to the former one. Concerning the vector $D^{-\frac{1}{2}}z_4$, its relatively irregular form suggests that two previously mentioned vectors are indeed adequate to differentiate all three clusters. This signifies, that the normalized *Graph Laplacian* encompasses the information about the appropriate number of clusters within analyzed data. However, this cannot be stated regarding the *Graph Laplacian* L_G. Upon examining the form of its eigenvectors v_2, v_3 and v_4, one may infer the existence of not three, but four clusters with the first one only marginally separated from the others.

Similar conclusion can also be drawn from the plots of the eigenvalues. Considering the unnormalized *Graph Laplacian*, there are four eigenvalues that are distinctly smaller than the others, whereas in the normalized case, there are only three such eigenvalues. This example highlights the general observation that the *normalized spectral algorithm* is better suited for unevenly distributed data with varying density, since it evaluates the degree of similarity within the potential clusters from a locally-based perspective, contrasting with its unnormalized version.

On the other hand, in the second example, the utilized data consists of 1500 two-dimensional points deliberately generated to exhibit a non-convex circular shape. It comprises three circles nested within each other, each with increasing radius, number of observations, and distortion level. The innermost circle contains 200 points, the medium circle contains 500 points, and the outermost circle contains 800 points.

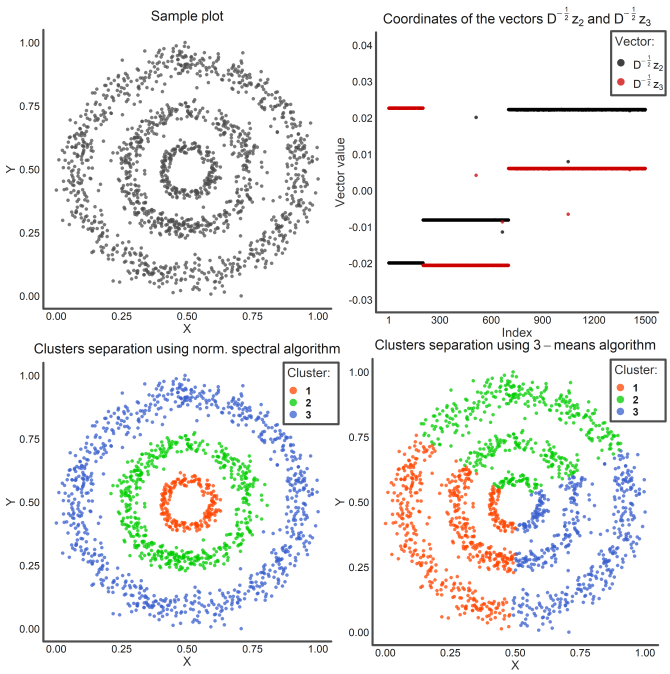

Figure 4.3: Data sample plot on the top left, subsequent entries of the vectors $D^{-\frac{1}{2}}z_2$ and $D^{-\frac{1}{2}}z_3$ plotted on the top right, clusters separation via the *normalized spectral algorithm* for 3 clusters on the bottom left, clusters separation via solely 3-means method on the bottom right.

In this example, for the *normalized spectral algorithm* to correctly detect distinct clusters, it was necessary to set a lower hyperparameter $\sigma = 0.01$ inside the *Gaussian similarity function*. Indeed, since nested circles are quite close to each other, to prevent the points forming them from being assigned to the same cluster, the algorithm should more strictly treat points as dissimilar, when the Euclidean distance between them increases.

Now, as one can see on the top right plot of Figure 4.3, the entries of the vectors $D^{-\frac{1}{2}}z_2$ and $D^{-\frac{1}{2}}z_3$ are easily distinguishable into three clusters. This signifies that, with such a *similarity measure*, the normalized *Graph Laplacian* captures all the necessary information on the resemblance within the data. In this way, the 3-means method applied to the distinct coordinates of these vectors has no problem identifying the clusters. In contrast, when applied to the initial data before the embedding, it fails to recognize the structure and shape of the data. This comparison is shown in the bottom plots of Figure 4.3.

These two described examples briefly illustrate the functioning and possibilities of the *spectral clustering algorithms*, but they also highlight the need for further insights into their practical aspects. Subsequent section will aim to address them in a more organized manner.

4.4. Practical aspects of spectral clustering

Finally, the success in accurately clustering data relies heavily on its examination and the appropriate adaptation of the entire process to the specific dataset. The spectral approach is no exception, therefore, to fully leverage its versatility, it is essential to rationally adjust the steps that contribute to this clustering method:

- **Choosing the similarity function** - This is the priority when it comes to grouping any type of observations, and an inappropriate choice of the *similarity measure* can lead to unsatisfactory results, regardless of the quality and effectiveness of the used algorithm. Most importantly, such a function should assign higher values to points that are considered similar to each other in the context of the data kind they come from, and lower values in the opposite case. That sounds quite obvious, however many times the critical line between similar and dissimilar is not so easy to establish throughout the entirety of the dataset. This is why the majority of *similarity functions* employ parameters that, when set adequately, make them more extensible to the complexity of the data. While working within the specific domain and shape, more direct recommendations on suitable *similarity function* and its parameters can be provided. For instance, when observations lie in the Euclidean space and their clusters have similar distributions and densities, a simple measure like the reciprocal of the distance should suffice to yield significant results. Conversely, more diverse data requires a function where a certain change in the distance between points leads to a more extreme change in their similarity value. This is precisely what can be achieved with the *Gaussian similarity function*: $s(x_i, x_j) = exp(\frac{\|x_i - x_j\|^2}{2\sigma^2})$, by experimenting with different parameters σ. Such flexibility makes it a fair default choice in the described circumstances.
- **Determining the similarity graph and the weighted adjacency matrix** - Right after the selection of the *similarity measure*, this step directly influences the form of the *Graph Laplacian*, which is the core of the *spectral algorithm*. In general, this involves choosing one of the four types of *similarity graphs* described in the Chapter 2, and then constructing the *weighted adjacency matrix* based on its form. In the case of *k-nearest neighbours graph* or

the ϵ-*neighbourhood graph*, once again setting their parameters adequately to given data plays a crucial role. Using these two *similarity graphs* gives strong consistency results, however, algorithms based on them may struggle when working on datasets with relatively close regions of different densities or a significant presence of outliers. In such cases, a more suitable option seems to be the *connected similarity graph*, which yields better results at the expanse of complexity and computational time throughout the whole process. Given the *similarity graph*, determining the *weighted adjacency matrix* typically involves assigning the respective similarities to the points, which are connected on a *similarity graph* and assigning zeros, otherwise.

- **Choosing the Graph Laplacian** - This part is definitely easier than the previous ones, as the main choice focuses only on the two types of *Graph Laplacians*. By examining their construction, it is quite evident that when all points have comparable *degrees*, both *Graph Laplacians* are almost identical with accuracy to scaling. Therefore, in such situation, the selection between them barely impacts the entire algorithm. However, when there are regions of high density along with sparse ones within the data, the normalized *Graph Laplacian* is recommended to effectively differentiate between these regions. Obviously, choosing the *Graph Laplacian* is equivalent to choosing the type of *spectral algorithm* that will be further executed.
- **Setting the number of potential clusters k** - It may seem like this aspect is a downside of the *spectral algorithm*, as the number of clusters is a mandatory parameter that needs to be specified initially for the algorithm, unlike some clustering methods that determine the appropriate number by themselves. Nonetheless, mentioned issue can be overcame with little effort by analyzing the eigenvalues of the chosen *Graph Laplacian*. Indeed, as shown in the second example in Section 4.3, the number of the first eigenvalues that were relatively smaller than the others corresponded to the proper number of clusters in the data. Relying on this gap is more of a heuristic approach than the analytic one, but it often yields adequate results.
- **Obtaining the clusters via the k-means method** - The final step of the algorithms entails transforming real-valued vectors associated with the respective *Graph Laplacian* into clusters within the data. When the original clusters are well-separated, the entries of these vectors are approximately partially constant, as depicted in the top right plot of Figure 4.3. Therefore, grouping them properly should not pose a challenge for any clustering method, particularly for k-means. Whereas, when the data comprises closely positioned or even intersecting clusters, these vectors exhibit a more curved form, leading to a wider margin of error for the k-means algorithm when attempting to differentiate the clusters. In such cases, the weighted k-means method can yield more desirable results by ensuring that the new centroids represent the truly central points within the clusters rather than those located close to their intersections.

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